A Self-Decryption Pass Transistor Logic-Based In-MRAM Computing Macro Using Hybrid VGSOT-MTJ/GAA-CNTFET

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***Abstract*— Spintronic devices and gate-all-around carbon nanotube field-effect-transistors (GAA-CNTFETs)-based computing in-memory architecture are competitive candidates for applications in battery-powered tiny artificial intelligence (AI) edge devices. Meanwhile, data encryption and decryption are also necessary to protect AI model weights and the customized data used to guarantee neural network (NN) inference accuracy. In this study, we propose a self-decryption pass transistor logic (PTL)-based in-MRAM computing macro (SP-CIM) that utilizes hybrid voltage-gated spin-orbit torque magnetic tunnel junctions (VGSOT-MTJ)/GAA-CNTFET. The proposed SP-CIM macro enables simultaneous data access, decryption, and full-accuracy multiply-and-accumulate (MAC) operations using the newly introduced voltage-divider self-decryption cell, without the need for additional decryption logic. Compared to existing in-memory decryption strategies, this design reduces energy consumption by 45.7% and decreases decryption delay by 87.2%. To enhance area efficiency and reduce computing latency, we propose a PTL-based multiplication cell that achieves full-accuracy local 2b-IN × 2b-W operations with only 20 transistors (20T). Additionally, novel PTL-based full-swing output half adders (10T-HA) and full adders (14T-FA) are proposed to construct the local adder tree, achieving reductions of 31.8%, 76.4%, and 41.4% in energy, delay, and area, respectively, compared to conventional adder trees in CIM macros. Simulations of the 288 kb SP-CIM macro demonstrated throughput and energy efficiency of 2.25 TOPS and 226.6 TOPS/W, respectively, at a 0.6 V supply voltage, and 2.97 TOPS and 154.1 TOPS/W, respectively, at a 0.8 V supply voltage, with 8b-IN, 8b-W, and 24b-OUT.**

***Index Terms*— Self-decryption, computing in-memory (CIM), pass transistor logic (PTL), PTL-based multiplication cell, PTL-based full swing output adder, GAA-CNTFET, VGSOT-MTJ, MRAM.**

Received 10 December 2024; revised 25 March 2025 and 24 April 2025; accepted 25 April 2025. Date of publication 12 May 2025; date of current version 27 October 2025. This work was supported in part by the Research Start-Up Funds of Hangzhou International Innovation Institute of Beihang University under Grant 2024KQ052; in part by Beijing Natural Science Foundation under Grant 4232070; in part by the National Natural Science Foundation of China under Grant 52261145694, Grant 62401031, and Grant 52402167; in part by the International Mobility Project under Grant B16001; in part by the Fundamental Research Funds for the Central Universities of China; in part by the Post-Doctoral Fellowship Program of China Postdoctoral Science Foundation (CPSF) under Grant GZC20233368; and in part by the Post-Doctoral Research Funding of Hangzhou International Innovation Institute of Beihang University under Grant 2024BKZ004. This article was recommended by Associate Editor X. Fong. *(Zhongzhen Tong and Sifan Sun contributed equally to this work.) (Corresponding authors: Xiaoyang Lin; Ke Zhang.)*

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Digital Object Identifier 10.1109/TCSI.2025.3566232

I. Introduction

As ARTIFICIAL intelligence (AI) models scale, their escalating computational and storage requirements intensify von Neumann architecture limitations, particularly energy inefficiency caused by frequent data transfers between processors and memory [1], [2], [3]. Computing in-memory (CIM) architecture allows direct calculations in memory, reducing the need for excessive access to storage during calculation processes [1], [2], [4], [5], [6]. Nevertheless, the rapid evolution of CIM architectures will introduce new security vulnerabilities [7], [8], especially for non-volatile memory e.g., MRAM [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19] and RRAM [20], because the stationary weights of AI models stored in memory potentially pose threats to inference engines, including model leaking and reverse engineering [13], [21], [22]. Generally, a well-trained AI model is valuable owing to the significant resources and time costs required for training [23]. Therefore, investigating a novel CIM macro to safeguard AI models is imperative.

As the CMOS process continues to scale, short-channel effects in silicon-based transistors are becoming increasingly significant [9], [24], [25]. This has led traditional CIM architectures based on silicon transistors to approach their energy efficiency limits [3], [23], [26]. In contrast, gate-all-around carbon nanotube field-effect transistors (GAA-CNTFETs) offer better gate control, resulting in reduced leakage currents and higher switching ratios (I_{ON}/I_{OFF}) [27], [28]. Both n-CNTFETs and p-CNTFETs feature symmetrical transfer characteristic curves, and the threshold voltage of CNTFET can be adjusted by the flat-band voltage (V_{fb}). Moreover, voltage-gated spin-orbit torque magnetic tunnel junction (VGSOT-MTJ) exhibits lower write energy and a read-write separated structure, resulting in higher endurance than RRAM [29]. Additionally, their integration allows for

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multi-layer monolithic 3D integration (a low-temperature, back-end-of-the-line-compatible process) [30], [31], superior radiation tolerance [32], [33], ternary system implementation [24], and excellent electrical properties such as high speed and low power consumption [4], [26], [34], [35]. Therefore, integrating CNTFET and MTJ technologies presents a promising path for constructing ultra-efficient CIM macros in the post-Moore era [9], [24], [25], [35]. Current research on hybrid CNTFET and MTJ designs for CIM primarily focuses on two aspects:

*1) Bit-cell level:*Current research on bit-cell level exploration includes ternary non-volatile memory cells for high-capacity storage [36], low-power true random number generators [37], near-memory logic computation units [4], approximate full adders [38], spintronic synapses and CNTFET neurons [34], [35], [39], and XNOR/XOR operation units for accelerating binary neural networks (BNNs) [26]. However, these studies are restricted to bit-cell level exploration, which imposes limitations on both throughput and energy efficiency.

2) Array level: In [9], Tong et al. developed a 16 kb MRAM array based on GAA-CNTFETs, enabling full-array activation for in-situ storage Boolean logic operations, which demonstrated high throughput in half-adder operations. However, its reliance on MTJ write operations (susceptible to thermal noise) for logic introduces significant power consumption and delay during arithmetic logic unit operations. In [25], an 8 kb analog BNN accelerator was implemented using a complementary-sensing time-based readout circuit. Although this design improves the read margin, it comes at the cost of increased computing delay. Moreover, it only supports binary multiply-and-accumulate (MAC) operations, making it unsuitable for high-precision computing scenarios. In contrast, [24] proposed a multi-bit ternary MAC CIM architecture that achieves high-density storage and efficient computation. Nevertheless, its 1b input (1b-IN) bit-serial strategy necessitates eight cycles for 8b input calculations, significantly increasing computing delay. Additionally, the reliance on local 1b×1b dot-product operations places a heavy burden on the subsequent adder tree. The CMOS-like ternary logic circuits (e.g., ternary adder tree and shifter-and-adder) further constrain area efficiency. Crucially, these MTJ-based CIM solutions universally lack mechanisms for AI model protection.

Therefore, to overcome the aforementioned issues, this study introduces a self-decryption pass transistor logic (PTL)-based in-MRAM computing macro (SP-CIM) using a hybrid VGSOT-MTJ [29] and GAA-CNTFETs [27], [28] for secure convolutional NNs (CNNs) acceleration. The advantages of the proposed SP-CIM are as follows.

1) We propose a voltage-divider self-decryption cell (VSDC) comprising four GAA-CNTFETs and two VGSOT-MTJs (4T2M), which enables simultaneous data access, decryption, and MAC operations without incurring additional decryption delay or circuit overhead.
2) A pass transistor logic (PTL)-based multiplication cell (PTL-MC) is proposed to achieve high area efficiency, low computing latency, and full-accuracy local 2b-IN×2b-W operations, using only 20 transistors (20T).
3) In the proposed SP-CIM, novel PTL-based full-swing output half adder (10T-HA) and full adder (14T-FA) are proposed to construct a local adder tree with 4b-IN/144 accumulation lengths. This leads to reductions of 31.8% in energy, 76.4% in delay, and 41.4% in area compared to existing adder trees used in CIM macros. Additionally, the proposed 10T-HA and 14T-FA are also utilized in the shift-and-add process to further enhance efficiency.

The remainder of this paper is organized as follows: **Section II** presents the challenges in existing CIM macros. **Section III** presents the overall structure, decryption schematic of VSDC, memory operations, operating principles of the proposed SP-CIM, and applications of CNNs. **Section IV** validates the performance of the proposed SP-CIM, provides a system benchmark for CNNs, evaluates the layout, compares the results with those of other studies, and discussing. Finally, **Section V** summarizes the study.

II. Challenges in Existing CIM Macros

A. High Overhead for Data In-Memory Decryption

In conventional von Neumann architectures, securing NN models requires a three-stage process: encrypted weights are retrieved from memory, decrypted via XOR operations with a Key (involving dedicated logic circuits), and then utilized for MAC operations. While pipelining mitigates latency, large-scale AI model protection introduces significant decryption latency, energy, and area overheads [21].

Volatile memory solutions: In [8], 8T SRAM cells with in-situ XNOR gates enable weight decryption operations. However, the 8T cell + 10T XNOR logic significantly compromises storage density and throughput. In [21], a 6T SRAM-based pre-computation scheme embeds in-memory XOR operations between the Key and IN at the wordline (WL) driver, achieving decryption access and analog MAC. Yet, limited analog computing margin and high ADC overhead degrade precision and efficiency.

Nonvolatile memory approaches: Chiu et al. [13] proposed a bidirectional bitline (BL)-access readout (BiBL) scheme where BL charge-discharge cycles synchronize self-decryption with read access for original/inverted data pairs. However, this method incurs high energy, delay, and area costs due to BL switching circuits (12T) and sense amplifier (SA) (28T), while mid-point reference sensing further limits read margin. Crucially, MAC operations remain sequentially dependent on decryption operations, constraining throughput.

As illustrated in **Fig. 1(a)**, our design employs a 4T2M voltage-divider self-decryption cell (VSDC) and a 2T inverter to achieve in-memory decryption with robust read margin. Moreover, data access, decryption, and MAC operations are executed simultaneously, significantly enhancing computing efficiency and throughput, compared to [13].

B. Large Logic Circuits Overhead in Existing CIM Macro

The primary idea of the digital CIM (DCIM) is to incorporate logic units into memory to execute MAC operations, which occupy a relatively large proportion of the area overhead. For example, Yi et al. integrated an adder tree using 26T mirror adders [40] while Fujiwara et al. employed 24T polarity-inverted adders [41] to design adder trees in SRAM arrays for MAC operations. However, these adder trees still impose considerable overhead. To reduce the logic circuit overhead, pass transistor logic (PTL), as shown in **Fig. 1(b)**, can replace CMOS-based logic, resulting in nearly a twofold reduction in T_{NUM} [42], [43], [44], [45], [46], [47], achieving

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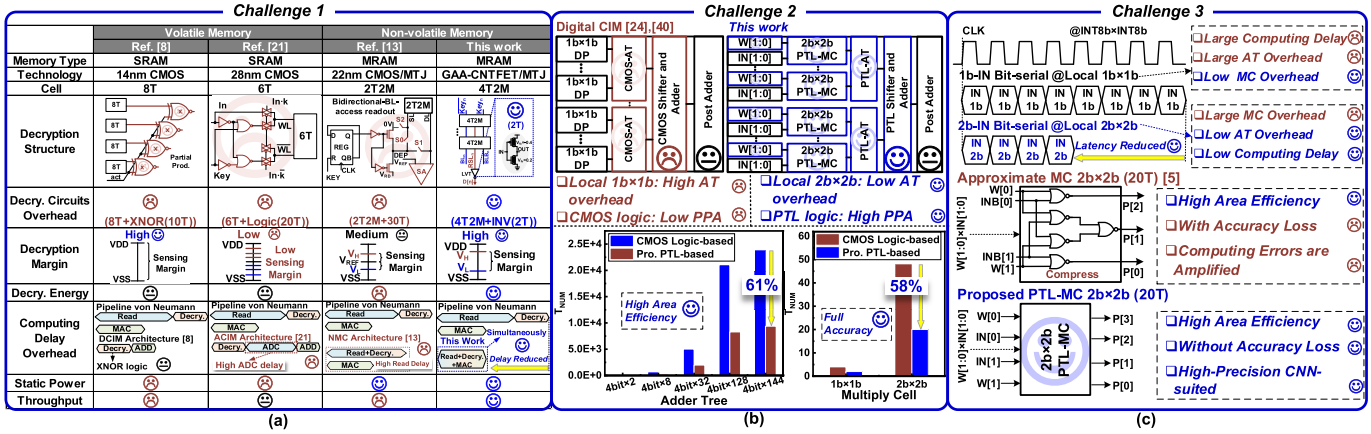


Fig. 1. Challenges in existing CIM macros: (a) High overhead for data in-memory decryption. (b) Large overhead for logic circuits in existing DCIM macros. (c) High overhead for input bit-serial CIM strategy.

high performance, power and area (PPA). Despite this advantage, PTL suffers from threshold voltage losses, leading to incomplete full-swing outputs, and long chains of transistors in series can cause incorrect results during adder tree operations in DCIM macros. For instance, in [42], Chih et al. alternated between 14T PTL-based full adders (with threshold losses) and 28T CMOS-based full adders to construct adder trees, aiming to save dynamic power and minimize excessive transistors in series. Similarly, Wang et al. introduced buffers (including two inverters) between every two 12T PTL-based full adders in an adder tree to mitigate issues with transistors in serial, reducing both area and power consumption [43]. However, these approaches add extra logic overhead to PTL-based adders.

To further enhance area and energy efficiency, in this study, based on a 6T PTL-based half adder (6T-HA), we propose PTL-MC realizing full-accuracy local 2b-IN×2b-W with only 20T. Also, full-swing output PTL-based 10T-HA and 14T-FA are proposed to achieve an adder tree with 4b-IN/144 accumulation lengths, where both the sum and carry are generated using inverters, addressing the issue of transistor in-serial leading to reduced signal drive capability, as detailly discussed in **Section III-D**. Compared with CMOS-based logic circuits, the proposed local adder tree and PTL-MC realized 61% and 58% decrease in transistor number (T_{NUM}).

C. High Overhead for Input Bit-Serial CIM Strategy

To reduce the overhead of multiplication circuits in CIM macros, many researchers have adopted the input bit-serial strategy [5], [14], [24], [41], [42], which generates partial product sums sequentially based on the input. For example, in a 1b-IN bit-serial scheme, an 8b-IN requires eight cycles [24], [41], [42]. The benefit of this approach is that it only needs local NOR (4T) logic to perform dot-product calculations [41], [42]. However, this strategy can place significant overhead on the adder tree when executing high-precision computations. In contrast, a 2b-IN bit-serial can achieve the 8b-IN in just four cycles [5], as shown in **Fig. 1(c)** thereby reducing computation latency. Nevertheless, this strategy incurs considerable logic circuit overhead when performing local 2b-IN × 2b-W calculations. To address this, Guo et al. proposed a 20T approximate multiplication cell [5], which improves area efficiency. However, when the input is "11" and the weight is "11," the result is "111" instead of the correct result "1001," leading to errors that can be amplified during the shift-and-add process [6], [24], ultimately degrading the inference accuracy in CNNs. In this work, we propose a self-decryption 2b-IN bit-serial CIM strategy based on the proposed PTL-MC and VSDC, offering low computation latency, high area efficiency, and error-free operations.

III. Proposed SP-CIM For Memory And CIM Modes

This section provides an overview of the proposed SP-CIM macro and describes the decryption schematic of VSDC, memory mode, self-decryption MAC operations, and applications for CNNs.

A. Macro Structure and Overview

Fig. 2 illustrates the overall structure of the SP-CIM macro. This macro consists of a 32 × 64 × 144b in-MRAM computing array, multilevel decoder, WL and BL drivers, Key storage circuits, SAs, and output circuits. The 32 × 64 × 144b in-MRAM computing array is split into 32 MRAM subarrays. Each MRAM subarray contains four self-decryption multiplication units (SDMUs). Each SDMU consists of 16 × 144b VSDCs, 144 decryption units, 144 PTL-MCs, and a PTL-based local adder tree with 4b-IN/144 accumulation lengths. Each VSDC comprises two VGSOT-MTJs and four GAA-CNTFETs. An 18b shifter-and-adder tree designed by PTL and post-adder are also included in the MRAM subarray. SP-CIM can operate in memory and self-decryption CIM modes that can be configured to support 4b-W/4b-IN and 8b-W/8b-IN MAC.

B. Decryption Schematic of VSDC

Fig. 3(a) presents the schematic of the VSDC adopting GAA-CNTFET proposed in [27] and [28], where M1–M4 are access transistors, and T1–T3 are the three terminals for the VGSOT-MTJ. The spin–orbit torque current (I_{SOT}) can be generated by the spin-Hall effect current, which is applied between T2 and T3. The MTJ can be switched from R_H (R_L) to R_L (R_H) without an external magnetic field, where R_H (R_L) is the high (low) resistance state of the MTJ. Additionally, the energy barrier can be manipulated using the voltage-controlled magnetic anisotropy effect (V_b), which substantially lowers

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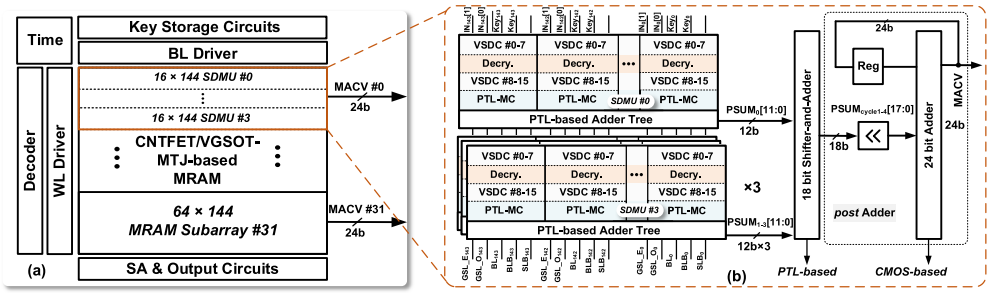


Fig. 2. Overall structure of the SP-CIM macro.

write energy consumption [29]. In VSDC, the WLR and RWL control the gates of M1 and M4, respectively, for read/write and self-decryption CIM operations. The RWR and WWL control gates M2 and M3, respectively. The SL and RSL are connected to drains of M3 and M4, respectively. BL and BL bars (BLB) can achieve read/write operations and can also be mapped as Key data. For instance, as illustrated in **Fig. 3(b)**, during the decryption operation, WLR, RWR and RWL are set to VDD. Then, Key_n (Key_0) is loaded onto BL_n/BLB_n (BL_0/BLB_0), thus, the column-wise decryption is carried out. The decryption results of VSDC are sensed by INV with a low threshold (LVT-INV), where the V_{fb} of p-CNTFET and n-CNTFET are −0.4 and 0.2 V, respectively obtaining a 0.3V threshold voltage (V_{LVT} = 0.3 V). The encrypted weight ($W^e = W \oplus Key$) is represented by MTJ_1 and MTJ_0, where W is the original weight. **Fig. 3(c)** shows the XOR value mapping table and decryption results, where BL/BLB = VDD/VSS (VSS/VDD) represents Key = 0 (1), MTJ_1/ MTJ_0 = R_L/R_H (R_H/R_L) represents W^e = 0 (1). The decryption operation results are reflected in the voltage of RSL (V_{RSL}). For example, if the Key and W^e are the same, V_{RSL} is as follows:

$$V_{RSL} = V_H = \frac{R_{M2} + R_H}{R_{M1} + R_{M2} + R_H + R_L}(VDD - V_{TH}) > V_{LVT}, \quad (1)$$

where V_H denotes that the computation result is greater than V_{LVT}, the V_{TH} is the threshold voltage of M1 or M2, and R_{M1} and R_{M2} are on-resistance (R_{ON}) of M1 and M2. Notably, R_H and R_L of VGSOT-MTJ are roughly 992kΩ and 336kΩ with a 200% tunnel magnetoresistance (TMR), respectively, while R_{M1}/R_{M2} is about 3.3 kΩ. The effects of VGSOT-MTJ resistance fluctuations will be detailed in **Section IV-A**. Thus, for simplification, we can ignore the R_{ON}. If the Key and W^e are different, then V_{RSL} will become:

$$V_{RSL} = V_L = \frac{R_L}{R_H + R_L}(VDD - V_{TH}) < V_{LVT}, \quad (2)$$

where V_L indicates that the computation result is less than V_{LVT}. Hence, the decryption operation between Key and W^e ($W^e \oplus Key$) can be read from LVT-INV.

The VSDC implements decryption through a voltage-divider mechanism, necessitating current flow through both MTJs during self-decryption operation. As shown in **Fig. 3(d)**, the adoption of VGSOT-MTJs in the VSDC achieves a 22.9–33.5× reduction in power consumption compared to [48], [49], and [50], a benefit directly attributed to the high-resistance characteristics of VGSOT-MTJs.

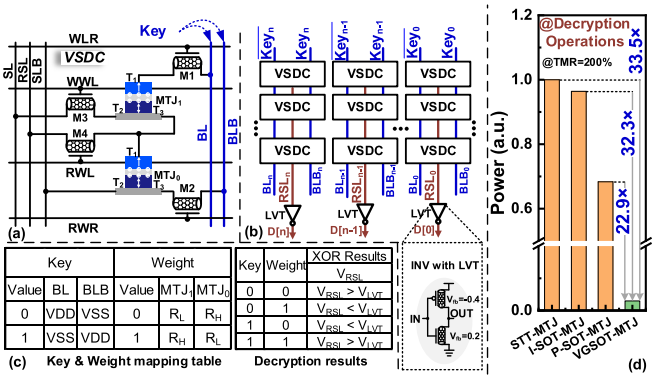


Fig. 3. (a) Schematic diagram of the VSDC. (b) The Keys are loaded onto the BLs/BLBs within the MRAM array. (c) XOR value mapping table and decryption results. (d) Self-decryption power comparison among various MTJs.

Consequently, the SP-CIM architecture leveraging VGSOT-MTJs enables energy-efficient self-decryption computing.

C. Memory Operations

1) Write Operations: The writing operations for MTJ_1 is illustrated in **Fig. 4(a)**, with MTJ_0 remaining idle. **Phase 1**: WLR, WWL, and RWL are activated. When SL is at V_{WR} (∼200mV) and RSL is at VSS, I_{SOT} flows from T2 to T3 to write R_L to MTJ_1. Conversely, the R_H is written to MTJ_1. At this time, BL is set to VDD, achieving $V_b > 0$ and reducing the minimum write current. **Phase 2**: RWL is then turned off, while WWL and WLR remain enabled. SL and BL are set to VDD and VSS, respectively, realizing $V_b < 0$ to stabilize the write operation. Similarly, the writing operations for MTJ_0 is shown in **Fig. 4(b)**.

2) Read Operations: **Fig. 4(c)** shows the read operation process. In the proposed SP-CIM, the drains of M4 in VSDC#0-7 (#8-15) are connected to RSL_E (RSL_O) while all drains of M1 are connected to BL. Thus, MTJ_1 can be read from BL. In contrast, MTJ_0 in VSDC#0-7 (#8-15) needs to be read from GSL_E (GSL_O). To accommodate different read operations, we use a MUX for switching. For example, to read MTJ_1 in VSDC, first, setting Ctrl.[1:0] to "10" establishes a connection between BL and SA, and BL is pre-charged to 0.5VDD. Next, GWL[1], WLR, and WWL are turned on. At this time, SL is set to VSS, and SLB is in a floating state; MTJ_0 is kept idle by turning off RWL and RWR. This allows BL to discharge from M1, MTJ_1 and M3 to VSS. Finally, when a sufficient voltage difference is established at BL, SA is activated to compare with V_{REF} achieving read operations. Similarly, MTJ_0 in

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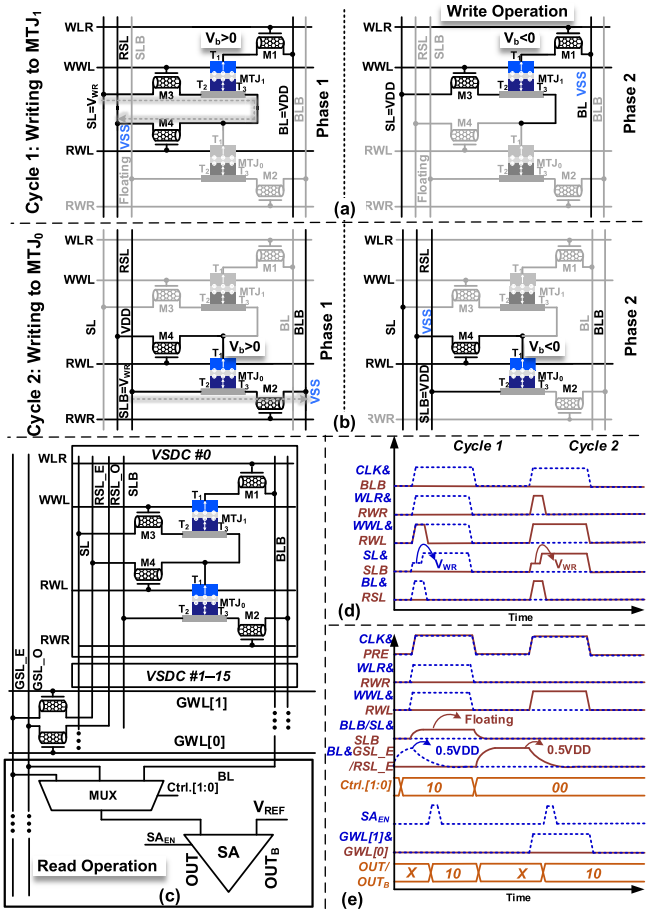
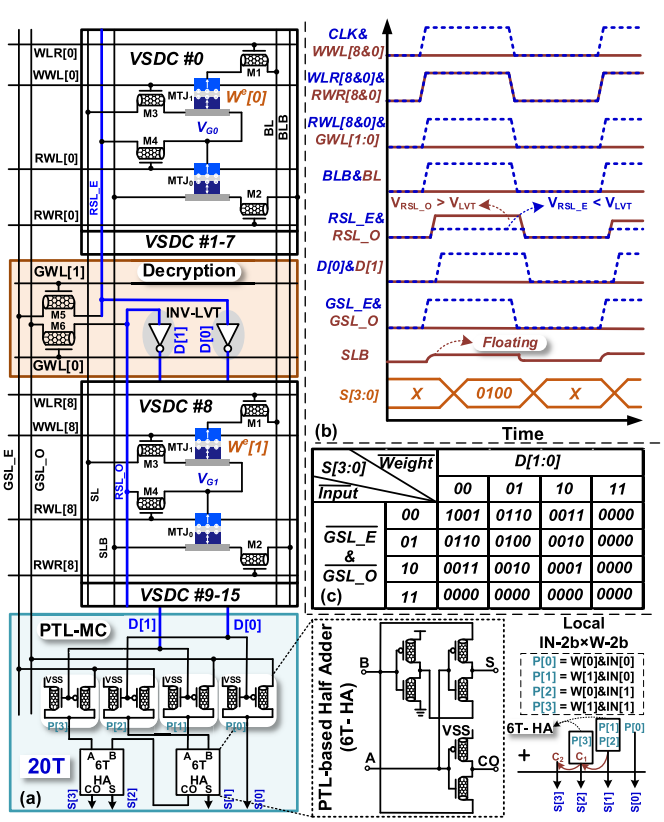


Fig. 4. Write operations: (a) Writing to MTJ1; (b) Writing to MTJ0. (c) Read operations. Transient simulation results of (d) write and (e) read process.



S[3:0] / Input	Weight	D[1:0] 00	01	10	11
$\overline{GSL_E}$ & $\overline{GSL_O}$	00	1001	0110	0011	0000
	01	0110	0100	0010	0000
	10	0011	0010	0001	0000
	11	0000	0000	0000	0000

Fig. 5. (a) Schematic of the self-decryption multiplication units. (b) Timing diagram for self-decryption multiplication operation with 2b-IN = "10" × 2b- W = "10". (c) Truth table for 2b-IN × 2b-W.

VSDC#0-7 (#8-15) can be accessed by setting Ctrl.[1:0] to "00" and "01", respectively. The timing sequences for the write and read operation are illustrated in **Figs. 4(d)** and **(e)**, respectively.

D. Self-Decryption CIM Operations

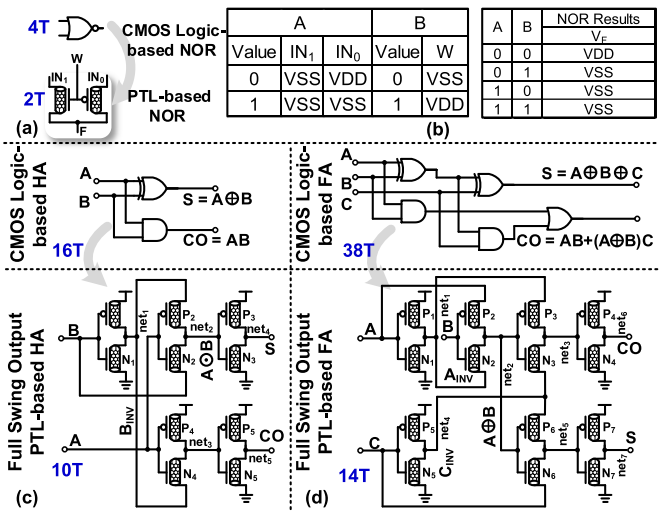
Fig. 5(a) presents a detailed block diagram of a 16 × 1 VSDC-based MRAM array for self-decryption multiplication. When performing self-decryption multiplication, $GWL[1:0]$ is set to VSS to isolate GSL_E/GSL_O from its neighboring RSL_E/RSL_O segments. The encrypted weights are stored in the array through the MRAM writing operations. To improve the throughput, two weight storage cells (VSDC in #0-7 and VSDC in #8-15) can be accessed simultaneously. The $\overline{Key}$ is loaded onto the BL and BLB. According to $\overline{Key}$, INV-LVT outputs the decrypted weight ($\overline{W[1:0]} = W^e[1:0] \oplus \overline{Key}$) and enters PTL-MC for multiplication operation. The 2b-IN ($\overline{IN[1:0]}$) are configured into GSL_E and GSL_O. Next, the $\overline{IN[1:0]}$ and $\overline{W[1:0]}$ are processed through the PTL-based NOR gate for the dot-product operation generating 4b partial products ($P[3:0]$). Finally, two 6T-HFs, as depicted in the lower right corner of **Fig. 5(a)**, are used to accumulate the $P[3:0]$ to obtain a partial product summation ($S[3:0] = IN[1:0] \times W[1:0]$). The proposed PTL-MC including 4 PTL-based NOR gates and two 6T-HFs, achieves full-precision local 2b×2b with only 20T, significantly enhancing area efficiency.

Take $W^e[1:0]$ = "10", $IN[1:0]$ = "10", and Key = "0" as an example. Firstly, in VSDC #8, the MTJ_1 and MTJ_0 are stored as R_H and R_L, respectively, representing $W^e[1]$ = "1"; Similarly, in VSDC #0, the MTJ_1 and MTJ_0 are stored as R_L and R_H, respectively, representing $W^e[0]$ = "0." BL and BLB are set to VSS and VDD, respectively, representing $\overline{Key}$ = "1." GSL_E and GSL_O are set to "1" and "0," respectively, indicating that the $\overline{IN[1:0]}$ is "01." $WLR[8\&0]/RWL[8\&0]$ and $RWR[8\&0]$ are simultaneously activated. $WWL[0]$ and $WWL[8]$ are set to VSS to turn off M3. Here, $V_{RSL_O} > V_{LVT}$ and $V_{RSL_E} < V_{LVT}$. Thus, the $\overline{W[1:0]}$ can be obtained from $D[1:0]$. Next, the $\overline{IN[1:0]}$ = "01" NORing $\overline{W[1:0]}$ = "01" operations are performed through 4 PTL-based NOR gates for dot-product operation. Finally, the $P[3:0]$ = "1000" are accumulated by two 6T-HFs to realize $S[3:0]$ = "0100." Therefore, the decryption and MAC operations can be completed simultaneously without additional decryption logic delays. The timing of the self-decryption multiplication operation is presented in **Fig. 5(b)**. The truth table of the multiplication results with 2b-IN and 2b-W is shown in **Fig. 5(c)**.

Notably, in **Fig. 5(a)**, the decryption circuit is placed between VSDC#0-7 and VSDC#8-15 to reduce the parasitic capacitance of RSL_O and RSL_E, thereby improving computation speed. This will be analyzed in detail in **Section IV-C**.

Fig. 6(a) shows that the T_{NUM} of the PTL-based NOR gate utilized in the proposed PTL-MC reduces from 4T to 2T compared with the CMOS-based NOR gate. The mapping and truth tables are shown in **Fig. 6(b)**. In addition, we propose

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A			B	
Value	IN_1	IN_0	Value	W
0	VSS	VDD	0	VSS
1	VSS	VSS	1	VDD

A	B	NOR Results V_F
0	0	VDD
0	1	VSS
1	0	VSS
1	1	VSS

Fig. 6. (a) The PTL-based NOR gate. (b) Mapping table and truth table for NOR logic. Proposed PTL-based full swing output (c) 10T-HA and (d) 14T-FA.

full-swing output PTL-based 10T-HA and 14T-FA, respectively, as shown in **Fig. 6(c)** and **(d)**. In [47], Kumar et al. proposed a 10T-HA, but it does not achieve full-swing output. A 14T full-swing output full adder was introduced in [45]; however, it utilizes three series-connected transmission gates for the Sum bit, which can result in significant signal transmission errors in adder trees with large accumulation lengths. Additionally, Yin et al. introduced an 18T-FA [46], while Pan et al. proposed a 16T-FA that requires extra inverters for ripple-carry adder (RCA) implementation [44]. Thus, utilizing the designs from [44] and [46] in the adder tree fails to deliver high area efficiency. Notably, because the Sum and Carry outputs of the proposed 10T-HA and 14T-FA are isolated by inverters, additional buffering is unnecessary to address PTL threshold loss in the implementation of the adder tree and shifter-and-adder, ensuring the high area efficiency and low energy consumption.

During the self-decryption CIM operation, two rows of data can be accessed, which means that multi-row WLs must be activated simultaneously. Therefore, we propose a multilevel decoder, as shown in **Fig. 7**. When executing memory mode, a traditional 6-64 decoder is utilized for reading and writing operations. In the CIM mode, a local decoder is designed that utilizes four 3-8 decoders. When the CIM signal is VSS, $Y[63:0]$ in the 6-64 decoder control $WL[63:0]$, correspondingly; when the CIM signal is VDD, the WLs are divided into four groups and controlled by four 3-8 decoders. The entire array is traversed via the $A_{CIM}[2:0]$ signal to complete computing operations.

The self-decryption MAC process with 8b-IN, 8b-W, and 24b-OUT is illustrated in **Fig. 8**, where the proposed PTL-based adder tree was designed using the proposed 10T-HA and 14T-FA with eight layers. In this adder tree, the least significant bit (LSB) of the RCA at each layer uses the proposed 10T-HA for reducing the T_{NUM}. Similarly, the 18b shifter-and-adder was built based on 10T-HA and 14T-FA with two layers.

The $W^e[7:0]$ are mapped to SDMU #0-3 via memory mode. For instance, the $W^e[1:0]$ ($W^e[7:6]$) are stored in SDMU#0 (SDMU#3). The $IN[7:0]$ are mapped to the

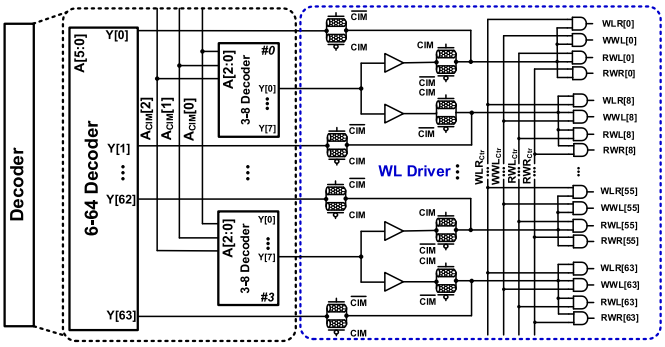


Fig. 7. Schematic diagram of the multilevel decoder and WL driver.

GSL_E/GSL_O in each cycle loading 2b-IN. Therefore, four cycles are required for the entire MAC operation.

In cycle 1, the $IN[1:0]$ is used for the calculations. In SDMU #0−3, the partial product summation ($S_0[3:0] - S_{143}[3:0]$) is accumulated by the PTL-based adder trees to output:

$$PSUM_{0-3}[11:0] = \sum_{i=0}^{143} IN_i[1:0] \times W_i[1:0]. \quad (3)$$

Next, $PSUM_{0-3}[11:0]$ are added with shifting 2b inputs, respectively, to realize:

$$PSUM_{cycle1}[17:0] = \sum_{i=0}^{143} IN_i[1:0] \times W_i[7:0], \quad (4)$$

where $PSUM_{cycle1}[17:0]$ denotes the 18b calculation results for cycle 1. The operation was repeated for three cycles, and the calculation result $PSUM_{cycle2-3}[17:0]$ was obtained for each cycle, loading 2b inputs. Since the 2b-IN are processed per cycle, in the post adder, the result of four cycles, $PSUM_{cycle1-4}[17:0]$, needs to be shifted by 2b for addition. Here the 24b registers used for storing the calculated results, a 24b adder designed by CMOS logic, and a 2b shifter are contained. Finally, the 24b MAC value ($MACV[23:0]$), is obtained from the post-adder as shown in **Eq. (5)**:

$$MACV[23:0] = \sum_{i=0}^{143} IN_i[7:0] \times W_i[7:0]. \quad (5)$$

The MAC operation with 4b-IN, 4b-W, and 16b-OUT is similar to the above process. Note that only two cycles are required to complete the operation. Additionally, the designed SP-CIM employs a 2b-IN bit-serial strategy, allowing it to reduce the cycle by half compared to a 1b-IN bit-serial [24], which significantly decreases computation latency.

E. Application of CNNs

In CNNs, the output feature (OF) is obtained by convolving the input feature (IF) with a kernel. The size of the convolution kernel is R × R × C × M, where R is the width/height, C is the number of input channels, and M is the total number of kernels. Similarly, the IF size is H × H × C, where H is the width/height and C is the number of channels. Convolution is a MAC operation in which the weights in the kernel are applied to the input feature, sliding from left to right and from top to bottom. The size of the OF (E × E × M) is determined by the size of the convolution kernel and the sliding stride size, where E represents the width/height and M represents the number of output channels.

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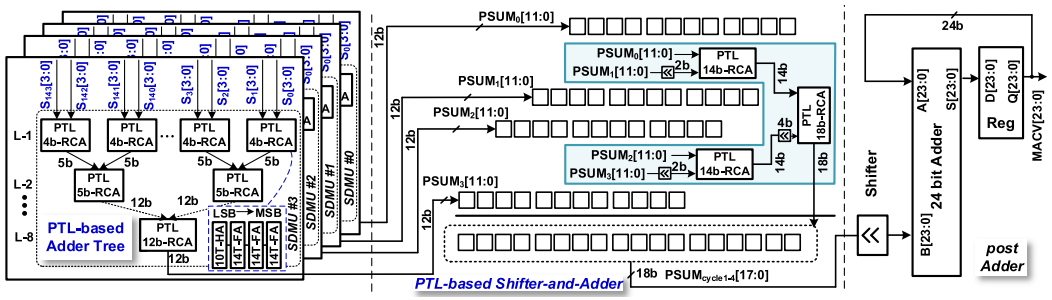


Fig. 8. MAC process with example of 8b-IN, 8b-W and 24b-OUT based on the proposed PTL-based adder tree, and PTL-based shifter-and-adder.

To protect the weights, the W^e is obtained by performing bit-wise XOR operations between the trained-weights and keys, as shown in **Fig. 9(a)**. The K-bit weights are divided into K/2 binary weights ($W_i^e[1:0]$ to $W_i^e[K-1:K-2]$). As shown in **Fig. 9(b)**, the K/2 binary weights are stored in SDMU #0−K/2. For example, the $W_{0,0}^e[1:0]-W_{0,N-1}^e[1:0]$ are stored in VSDC #0 and VSDC #8 by rows (presented in SDMU #0), where N is the number of weights in kernel #0. Similarly, $W_{0,0}^e[K-1:K-2]-W_{0,N-1}^e[K-1:K-2]$ are stored in VSDC #0 and VSDC #8 by row (presented in SDMU #K/2). The Keys are mapped to BLs/BLBs. The N binary K-bit inputs are mapped to GSL_E/GSL_O by loading 2b per cycle, where K/2 cycles are required to achieve a self-decryption MAC operation (from $IN_i[1:0]$ to $IN_i[K-1:K-2]$), as shown in **Eq. (6)**:

$$MACV=\sum_{i=0}^{N-1} IN_i[K-1:0]\times W_i[K-1:0]. \quad (6)$$

Encrypted kernel #j is similarly stored in SDMU #0-K/2 by row. Moreover, in the proposed SP-CIM, 32 MRAM subarray can be simultaneously activated to increase the throughput of the MAC operation.

IV. Simulation Results and Analysis

The proposed SP-CIM was simulated based on SPICE using hybrid VGSOT-MTJ [29]/GAA-CNTFET [27], [28], 14nm FinFET/VGSOT-MTJ and 28nm CMOS/VGSOT-MTJ. The important parameters of the GAA-CNTFET and VGSOT-MTJ devices are listed in **Table I** with inevitable process variations (including TT, SS, FF SF and FS corners) similar to [37]. In this study, 200% TMR was selected to achieve the proposed circuit design. The minimum gate width of the GAA-CNTFETs in the proposed SP-CIM CIM macro was 18 nm. The V_{fb} used in INV-LVT are -0.4 and 0.2 V, respectively; others are 0V. Additionally, to verify the robustness of the proposed SP-CIM, Monte Carlo simulations are also performed. The critical parameters of GAA-CNTFET and VGSOT-MTJ are assumed to follow a Gaussian distribution with 10% and 15% variations, respectively at the $\pm 3\sigma$ level [39], [51], as shown in **Table II**. The memory size was $32 \times 64 \times 144$ to obtain a 288 kb array.

A. The Proposed SP-CIM Performance Analysis

1) Basic VSDC Performance Evaluations: To analyze the stability of the self-decryption operations, we evaluated the V_{RSL} and sensing margin defined as $V_H - V_L$ under different TMRs, the results of which are shown in **Fig. 10(a)**. The larger the TMR of the VGSOT-MTJ, the larger the sensing margin.

TABLE I
Important Parameters of GAA-CNTFET and VGSOT-MTJ Model

	Parameters	Typical Value (T)		Corner Value	
				Slow (S)	Fast (F)
GAA-CNTFET	Transistor width (W)	18nm		Value-2nm	Value+2nm
	Physical gate length (Lg)	12nm		14nm	10nm
	Contact length (Lc)	12nm		Value-2nm	Value+2nm
	Source/drain extension length (Lext)	3nm		2.8nm	3.2nm
	CNT diameter (d)	1.2nm		1.1nm	1.3nm
	Gate oxide thickness (tox)	3nm		3.2nm	2.8nm
	Gate height (Hg)	20nm		Value-2nm	Value+2nm
	Flat band voltage (Vfb)	LVT	-0.4/0.2 V	Value+50mV	Value-50mV
		Others	0 V		
	Spacing between the CNTs (s)	3nm		3.2nm	2.8nm
VGSOT-MTJ	Diameter of MTJ (D)	50nm		48nm	52nm
	MgO barrier thickness (tox)	1.4nm		1.45nm	1.35nm
	TMR ratio under zero bias voltage (TMR_0)	200%		220%	180%
	MTJ surface length (a)	50nm		48nm	52nm
	MTJ surface width (b)	50nm		48nm	52nm
	MTJ surface radius (r)	25nm		23nm	27nm
	AFM-strip width (W)	50nm		48nm	52nm
	AFM-strip thickness (d)	3nm		3.2nm	2.8nm
	AFM-strip length (l)	60nm		58nm	62nm

TABLE II
Monte Carlo Parameters of the GAA-CNTFET and MTJ Model

	Monte Carlo Parameters	±3δ Variation
GAA-CNTFET	Physical gate length (Lg)	10%
	Contact length (Lc)	10%
	Gate height (Hg)	10%
	Gate oxide thickness (tox)	10%
	Flat band voltage (Vfb)	10%
	The number of CNTFET (Ncnt)	10%
MTJ	MgO barrier thickness (tox)	15%
	Free layer thickness (tsl)	15%
	TMR ratio under zero bias voltage (TMR_0)	15%

When TMR is 200%, 434 and 206 mV can be obtained for V_H and V_L, respectively. The sensing margin is 228 mV. This sensing margin can be easily distinguished using the INV-LVT. Thus, in this study, a 200% TMR was selected to achieve the proposed self-decryption CIM design. **Fig. 10(b)** illustrates the variations of R_H and R_L under 200% TMR, respectively. The results show that R_H (R_L) is 992kΩ (336kΩ) with a deviation of 33.75kΩ (1.42kΩ). Additionally, we simulated the R_{ON} of the GAA-CNTFET, which resulted in 3.3kΩ. Therefore, compared to the R_H and R_L, R_{ON} can be neglected in the decryption process. Moreover, the high resistance values of R_H and R_L can reduce the current in the VSDC, but also slow down signal establishment on RSL. To enhance computing speed, the decryption circuit is positioned between VSDC#0-7 and VSDC#8-16, effectively reducing the RSL parasitic capacitance, as detailed in **Section IV-C**. **Fig. 10(c)** illustrates the current duration during the decryption process

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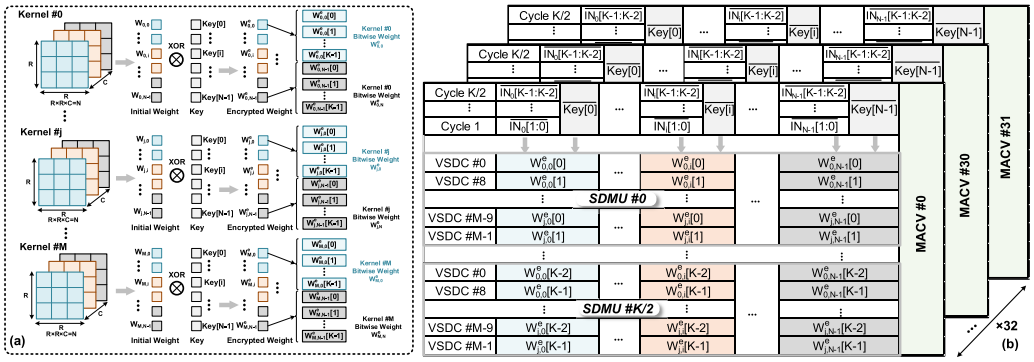


Fig. 9. (a) Bit-wise encrypting for weights. (b) Mapping weights, inputs, and keys into the proposed SP-CIM.

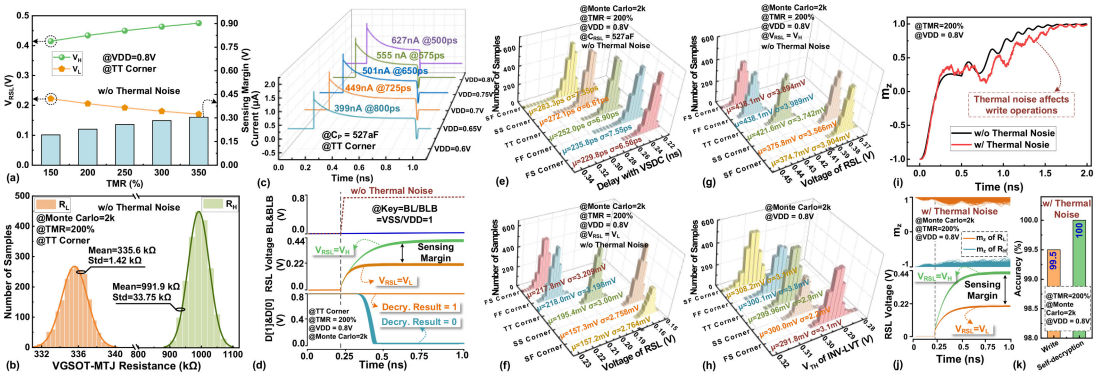


Fig. 10. (a) Simulation results of the VRSL and sensing margin with various TMRs. (b) The variations of RH and RL for VGSOT-MTJ at TMR = 200%. (c) The duration of the current during the decryption process within VSDC. (d) Monte Carlo simulation results for self-decryption operations. Variation analysis under different process corners: (e) Self-decryption delay within VSDC. Simulation results of RSL voltages with (f) VL and (g) VH. (h) VTH of INV-LVT. (i) Write operations comparison between w/o thermal noise and w/ thermal noise. (j) Monte Carlo simulation results for self-decryption operations w/ thermal noise. (k) Accuracy of write and self-decryption w/ thermal noise.

at various VDDs. The results indicate that when VDD is 0.8V (0.6V), the decryption current lasts for 500ps (800ps), which is sufficient for stable signal establishment.

2) Robustness Analysis: **Fig. 10(d)** presents Monte Carlo simulation for self-decryption results of "1" and "0." Due to sufficient read margin during the decryption process, the LVT-INV reliably distinguishes RSL voltage, ensuring computational accuracy. **Fig. 10(e)** shows the Monte Carlo simulation of computation delay in the VSDC across different process corners, where the delay is defined as the time from WL activation to the correct establishment of INV-LVT. In the TT corner, the computation delay is 252ps with a deviation of 6.9ps. Even in the SF process corner, the delay remains only 283ps, ensuring high computation speed. **Figs. 10(f)** and **(g)** illustrate the variations of V_L and V_H, respectively. Although V_L or V_H differs by nearly 65mV between the FS and SF corners. This does not impact the correct reading of INV-LVT. **Fig. 10(h)** shows the V_{TH} of INV-LVT across different process corners. The simulation results demonstrate that it can reliably distinguish the outcomes shown in **Figs. 10(f)** and **(g)**.

Thermal noise in VGSOT-MTJs impacts circuit operational reliability. **Fig. 10(i)** demonstrates the effect of thermal noise on write operations, where m_z denotes the MTJ magnetization state. The results show that thermal noise deteriorates write operation fidelity. **Fig. 10(j)** characterizes the impact of thermal noise on self-decryption processes, where voltage levels of V_H and V_L maintain sufficient sensing margins. As quantified in **Fig. 10(k)**, thermal noise increases write error rates by 0.5% but leaves self-decryption accuracy unaffected, validating the robustness of the proposed self-decryption scheme.

During the read operation, the BL voltage is transmitted to the SA input through the MUX. To assess the stability of the read operation, we simulated the SA input voltage when reading R_L and R_H, as shown in **Fig. 11(a)**. The results indicate a reading margin of 93.3mV with a 200% TMR. **Fig. 11(b)** displays the offset voltage of the SA under various process corners, with a maximum of 17.66mV in the FF corner. This value is below the reading margin, thus, ensuring the robustness of the read operation.

3) Energy and Delay for Memory and Self-Decryption CIM Operations: **Fig. 11(c)** shows the energy consumption and delay of the proposed SP-CIM for the memory mode. **Fig. 11(d)** shows the energy consumption and delay of the proposed SP-CIM for the CIM mode under different VDDs at TMR = 200%. When VDD = 0.6 and 0.8V, the energy consumptions of the MAC operations are 22.2 and 33.2 pJ, respectively, and the delays of the MAC operations are 2.1 and 1.5 ns with 4b-IN/4b-W/16b-OUT; The energy consumptions of the MAC operations are 40.6 and 59.8 pJ, respectively, and the delays of the MAC operations are 4.1 and 3.1 ns at VDD = 0.6 and 0.8V with 8b-IN/8b-W/24b-OUT. The MAC with 8b-IN/8b-W/24b-OUT can be completed in 3.1 ns (4 cycles) at VDD = 0.8 V, resulting in a high throughput.

4) Distributions of Delay and Energy: **Fig. 11(e)** shows the delay distribution in CIM, with the array, PTL-based adder tree, and post adder tree contributing 29%, 22.8%, and 19.6%, respectively, where the delay in array specifically

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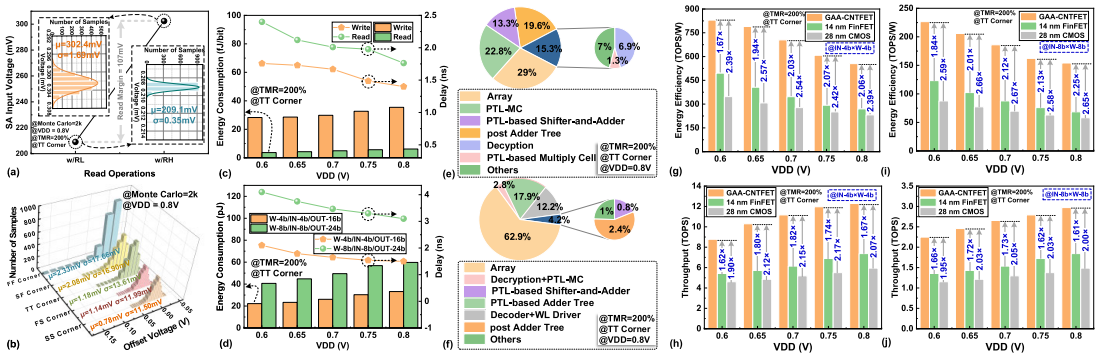


Fig. 11. (a) Read operations for SA input voltage with reading RL and RH. (b) SA offset voltage. Energy consumption and delay of (c) memory operations and (d) self-decryption CIM operations. Distributions of (e) delay and (f) energy. Simulated (g) energy efficiency and (h) throughput for SP-CIM at IN-4b/W-4b. Simulated (i) energy efficiency and (j) throughput for SP-CIM at IN-8b/W-8b.

represents the delay in establishing the RSL voltage within the VSDC. In addition, the energy distributions are shown in **Fig. 11(f)**. As can be seen, the array and PTL-based adder tree account for 62.9% and 17.9% of the total energy, respectively. Furthermore, the decoder and WL driver, and post adder tree accounted for 12.2% and 2.4%, respectively.

5) Energy Efficiency and Throughput Evaluations: For a rigorous comparative analysis of the proposed SP-CIM architecture, energy efficiency and throughput were benchmarked across three technologies: 12nm GAA-CNTFET, 14nm FinFET, and 28nm CMOS (**Figs. 11(g)–(j)**). Under IN-4b/W-4b, the GAA-CNTFET-based SP-CIM exhibits 1.67×–2.57× energy efficiency gains and 1.62×–2.17× throughput improvements over 14nm FinFET and 28nm CMOS implementations. These advantages scale further to 1.84×–2.67× (energy efficiency) and 1.61×–2.05× (throughput) under IN-8b/W-8b. The proposed 288 kb SP-CIM can perform 64 × 144 × 2 self-decryption MAC operations (OP) of 4b × 4b in 2 cycles, or 32 × 144 × 2 operations of 8b × 8b in 4 cycles, with multiplication and accumulation counted as two operations. At VDD = 0.6 V and VDD = 0.8 V, the energy efficiencies of GAA-CNTFET-based SP-CIM are 830 (calculated as 64 × 144×2OP/22.2 pJ) and 555 TOPS/W, and the throughputs are 8.78 (calculated as 64 × 144×2OP/2.1ns) and 12.29 TOPS, respectively, with 4b-IN/4b-W/16b-OUT (**Figs. 11(g)** and **(h)**). Further, at VDD = 0.6 V and VDD = 0.8 V, the energy efficiencies of GAA-CNTFET-based SP-CIM are 226 and 154 TOPS/W, and the throughput are 2.25 and 2.97 TOPS, respectively, with 8b-IN/8b-W/24b-OUT (**Figs. 11(i)** and **(j)**).

B. CNNs Benchmarks

CNNs including LeNet-5 and VGG-16 were implemented using the PyTorch package to validate the functionality and performance of the proposed SP-CIM in the hardware presentation of the NN. The LeNet-5 (VGG-16) model comprises 3 (13) convolution layers and 2 (3) fully connected layers, with 61.47K (15.24M) weight parameters and 416.5K (15.47G) MAC operations for identifying a given image in the MNIST (CIFAR-10) dataset. The energy consumption and delay of each layer depend on the number of MAC operations. **Figs. 12(a)** and **(b)** show the evaluation results, including the energy and delay per classification, using LeNet-5 on the MNIST dataset. Similarly, **Figs. 12(c)** and **(d)** present the evaluation results, including the energy and delay per classification using VGG-16 for the CIFAR-10 dataset. In this study, because no error is included in the self-decryption MAC operation, compared with the float-point network model, as shown in **Fig. 12(e)**, the proposed SP-CIM with 8b-IN/8b-W and 4b-IN/4b-W can achieve 98.92 and 98.86%, respectively, under the MNIST dataset and 93.33 and 92.27%, respectively, under the CIFAR-10 dataset, where the precision loss is due to quantifying the CNNs model from 32 floating-point to integer. Compared to [5], [16], and [17], the proposed SP-CIM avoids the issue of MAC operation errors in DCIM being amplified by the shift-and-add process [6], [24].

C. Layout Evaluation

Fig. 13(a) displays the layout of VSDC #7-8 and the decryption circuit based on the λ-based rule [11], [27], [28], as shown in **Fig. 13(b)**, where the λ is the minimum feature size for 12 nm GAA-CNTFET with 6nm. The proposed design requires the simultaneous reading of 2b-W, which necessitates the use of even-odd BLs, resulting in increased area overhead. To improve area efficiency, we employed a common area (CA) [11] technique that places BLs on different metal layers. For example, in the layout shown in **Fig. 13(a)**, there are 7 BLs: GSL_O, SLB, BL, and BLB are on metal-2, while GSL_E, RSL_E/O, and SL are on metal-3. Without using CA, there would be 2.33 times the additional area overhead compared to designs with CA and without even-odd BLs. However, with CA, the proposed even-odd BLs incur only 1.33 times the additional area overhead as shown in **Fig. 13(d)**.

Fig. 13(c) presents the layout of the mini-MRAM array including two 8 × 144 MRAM array and decryption circuits, which occupies an area of 84.8 μm^2. Notably, the decryption circuit was placed between the two mini arrays effectively reducing the half parasitic capacitance of RSL. Additionally, to improve the reliability of the simulation results, an estimated wire model from the NVSim platform [52] was utilized. The capacitance per unit length is 0.2724 fF/μm, with parasitic capacitances of 527 aF for the RSL (the length is 1.936 μm). Moreover, the parasitic capacitance of other BLs (GSL_E, GSL_O, SLB, BL, and BLB) and WLs (WWL, WLR, RWL, RWR and GWL[1:0]) were considered in the proposed SP-CIM simulations. Moreover, the parasitic capacitances of GAA-CNTFET including gate-to-CNT fringe and gate-to-contact coupling capacitances have been considered in [27] and [28].

The area of the proposed SP-CIM is clarified by the area distribution shown in **Fig. 13(e)**. The array and PTL-based

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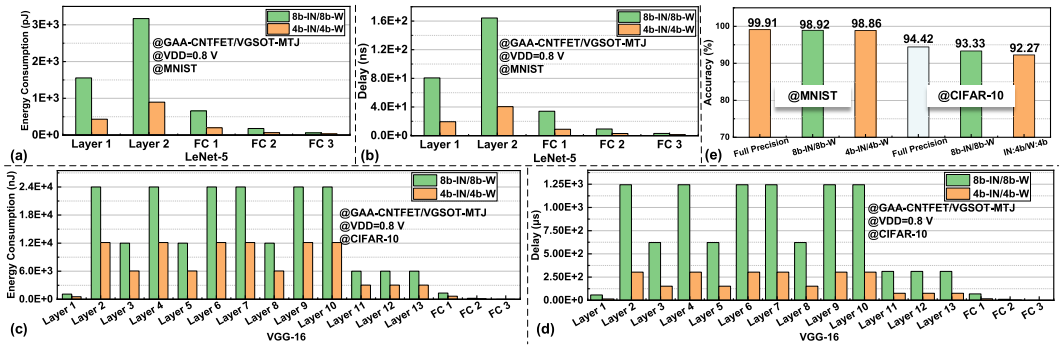
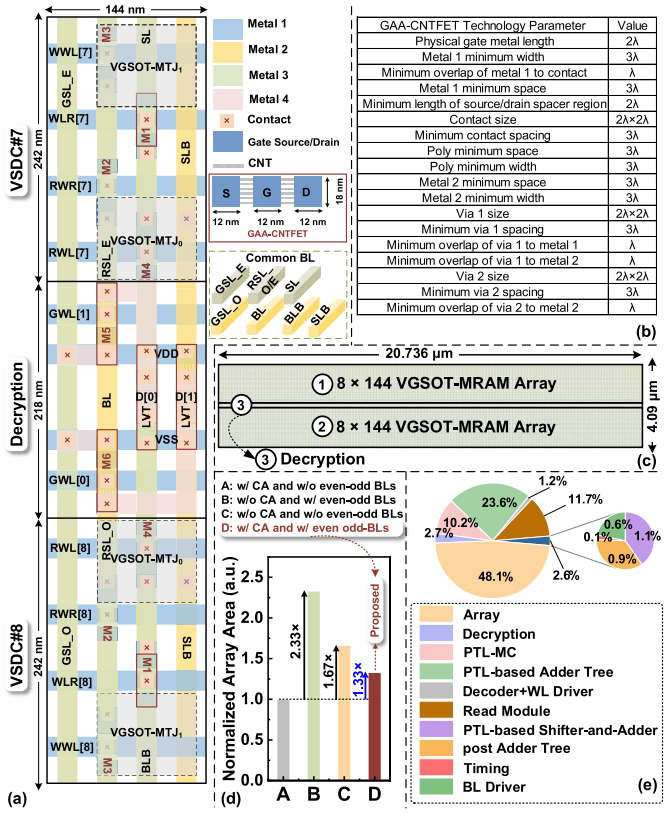


Fig. 12. Simulated performance of (a) energy and (b) delay for LeNet-5 applied to SP-CIM with MNIST dataset. Simulated performance of (c) energy and (d) delay for VGG-16 applied to SP-CIM with CIFAR-10 dataset. (e) Simulated accuracy with the MNIST and CIFAR-10 datasets.



GAA-CNTFET Technology Parameter	Value
Physical gate metal length	2λ
Metal 1 minimum width	3λ
Minimum overlap of metal 1 to contact	λ
Metal 1 minimum space	3λ
Minimum length of source/drain spacer region	2λ
Contact size	2λ×2λ
Minimum contact spacing	3λ
Poly minimum space	3λ
Poly minimum width	3λ
Metal 2 minimum space	3λ
Metal 2 minimum width	3λ
Via 1 size	2λ×2λ
Minimum via 1 spacing	3λ
Minimum overlap of via 1 to metal 1	λ
Minimum overlap of via 1 to metal 2	λ
Via 2 size	2λ×2λ
Minimum via 2 spacing	3λ
Minimum overlap of via 2 to metal 2	λ

Fig. 13. (a) The layouts of VSDC #7-8 and decryption circuits. (b) Layout design rules of the 12 nm GAA-CNTFET technology node [11], [27], [28]. The minimum feature size $\lambda = 6$ nm. (c) The layout of mini-MRAM array. (d) Array area comparison between w/ CA and w/o even-odd BLs, w/o CA and w/ evenodd BLs, w/o CA and w/o even-odd BLs, and w/ CA and w/ even-odd BLs. (e) Area breakdown.

adder tree account for 48.1 and 23.6% of the total area. The contribution of PTL-MC, PTL-based shifter-and-adder and post adder tree is estimated to be about 10.2, 1.1 and 0.9%, respectively. Additionally, for memory operations, the read module, decoder + WL driver and BL driver account for 11.7, 1.2 and 0.6% of the total area, respectively.

D. Comparison

In conventional in-memory decryption, the process consists of four steps: pre-charging, signal developing, sensing, and data decryption. Reference [13] introduced a

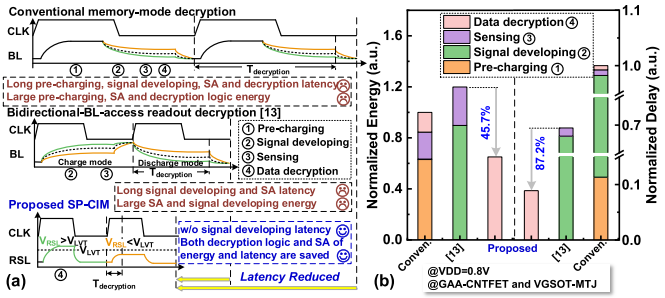


Fig. 14. (a) In-memory decryption process and (b) Normalized energy and delay comparison for conventional, bidirectional-BL-access readout decryption and the proposed SP-CIM.

bidirectional-BL-access readout decryption strategy, which eliminates the pre-charging and data decryption, reducing the decryption latency. However, as shown in **Fig. 14(a)**, the proposed SP-CIM macro using VSDC streamlines the in-memory decryption process into a single step, bypassing both pre-charging and signal developing. This results in significantly lower computing latency and energy consumption. To demonstrate the benefits of the proposed design in self-decryption, both the conventional approach and the method from [13] were implemented using GAA-CNTFET and VGSOT-MTJ technology to ensure a fair comparison. Simulation results reveal that the proposed design reduces energy consumption by 45.7% and delay by 87.2% compared to [13], as shown in **Fig. 14(b)**.

To highlight the advantages of the proposed PTL-based adder tree, we compared it with state-of-the-art adder trees [40], [41], [42], [43] used in CIM applications in terms of area, energy consumption and delay, as shown in **Fig. 15**. For a fair comparison, all the adder trees were implemented using GAA-CNTFET technology. **Figs. 15(a)** and **(b)** demonstrate the layouts of CMOS-based and the proposed PTL-based HAs and FAs. While PTL implementations require more complicate routing and exhibit reduced source/drain sharing between transistors, they achieve 30.1–57.1% area reduction compared to CMOS-based counterparts, primarily attributable to their lower T_{NUM}. For a 4b-IN adder tree with 144 accumulation lengths, the proposed PTL-based implementation achieves 27.9–41.4% area reduction compared to [40], [41], [42], and [43] (**Fig. 15(c)**). Additionally, simulation results show that the proposed PTL-based adder tree achieves up to 31.8% reduction in energy consumption and 76.4% reduction in delay, respectively compared to [40] and [43] (**Fig. 15(d)**).

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TABLE III

COMPARISON OF THE PROPOSED STUDY WITH THE PREVIOUS IN MEMORY COMPUTING STUDIES

	This work			ISSCC'22 [13]	TVLSI'23 [23]	TCAS-I'24 [40]	Nature'22 [18]	JSSC'24 [5]	ISSCC'24 [3]	ISSCC'23 [19]	TCAS-I'21 [16]
Technology	12nm GAA-CNTFET /VGSOT-MTJ	14nm FinFET /VGSOT-MTJ	28nm CMOS /VGSOT-MTJ	22nm CMOS/STT-MTJ	55nm CMOS	55nm CMOS	28nm CMOS /STT-MTJ	28nm CMOS	22nm CMOS	22nm CMOS /STT-MTJ	28nm CMOS /SOT-MTJ
Cell Type	4T2M			1T1M	6T	8T	3T2M	6T	6T	1T1M	2T1M
Array Size	288 kb			4 Mb	3 kb	64 KB	4 kb	64 kb	64 kb	8 Mb	1 MB
Supply Voltage (V)	0.6-0.8			0.8	1.2	1.2	0.8-1	0.6-0.9	0.6-0.9	0.65-0.8	1
Access Time[5] (ns)	1.5 [3,7] 3.1 [3,8]	2.5 [3,7] 5 [3,8]	3.1 [3,7] 6.2 [3,8]	9.8 - 22.1 [3,8]	17	NA	90.1 [13]	5.5 [8]	4.1 [17]	20 [3,8]	253.58
Input/Weight/ Output Precision	4b/4b/16b			8b/8b/26b	4b/4b/5b	4b/4b/NA	1b/1b/4b	8b/8b/23b	8b/8b/23b	8b/8b/26b	4b/4b/11b
	8b/8b/24b					8b/8b/NA					
Ideal Output Precision	16b/24b			26b	13b	NA	6b	23b	23b	26b	11b
Operation Mode	Digital CIM			Digital CIM	Analog CIM	Digital CIM	Analog CIM	Digital CIM	Digital CIM	Digital CIM	Digital CIM
Supported Decryption	Yes (XOR-cipher)			Yes (XOR-cipher)	Yes (XOR-cipher)	NO	NO	NO	NO	NO	NO
Throughput (TOPS)	8.78 [1,4,7]-12.29 [1,3,7]	5.42 [1,4,7]-7.37 [1,3,7]	4.61 [1,4,7]-5.95 [1,3,7]	0.26 [2,8]	1.32 [2,7]	0.81 [2,7]	0.09 [2,3,13,14]	0.375 [2,8]	0.062 [2,8,17]	0.59 [2,8]	NA
	2.25 [1,4,8] - 2.97 [1,3,8]	1.35 [1,4,8] - 1.84 [1,3,8]	1.15 [1,4,8] - 1.49 [1,3,8]			0.20 [2,8]					
Energy Efficiency (TOPS/W)	555 [1,3,7] - 830 [1,4,7]	270 [1,3,7] - 495 [1,4,7]	232 [1,3,7] - 347 [1,4,7]	25.1 [2,8] - 55.1 [2,8]	93.35 [2,7]	66.3 [2,7]	405 [2,3,14]	19.5 [2,8,17]	20.7 [2,8,17]	46.4 [2,3,8]	12.28 [1,7]
	154 [1,3,8] - 226 [1,4,8]	68 [1,3,8] - 123 [1,4,8]	58 [1,3,8] - 88 [1,4,8]			16.6 [2,8]		44 [2,4,8]			
FoM[6]	8888 [3,7]	4318.6 [3,7]	3708.6 [3,7]	3526.4 [8]	574.5 [7]	1060.8 [7]	270 [14]	1248 [8,17]	1324.8 [8]	2969.6 [3,8]	196.48 [7]
	9862.4 [3,8]	4383.3 [3,8]	3719.9 [3,8]			1062.4 [8]					
Accuracy in Neural Network (%)	98.86 [7,9]; 98.92 [8,9]			NA	97.6 [7,9]	94.1 [8,10,11]	98.86 [9]	74.51 [8,15]	80.03 [16]	NA	98.65 [9]
	92.27 [7,10]; 93.33 [8,10]					94.8 [8,10,12]					91.11 [10]

[1]Simulation; [2]Chip measurement; [3]Performance @VDD=0.8 V; [4]Performance @VDD=0.6 V; [5]The time required to complete one CIM operation; [6]Energy efficiency × input precision × weight precision × output ratio; [7]Performance @ 4b-IN/4b-W; [8]Performance @ 8b-IN/8b-W; [9]Performance @MNIST; [10]Performance @CIFAR-10; [11]Performance @ResNet-10; [12]Performance @ResNet-18; [13]Calculated through the operating frequncy (11.1MHz); [14]Performance @ 1b-IN/1b-W; [15]Performance @CIFAR-100; [16]Performance @Vision transformer for ImageNet; [17]Performance @VDD=0.9 V.

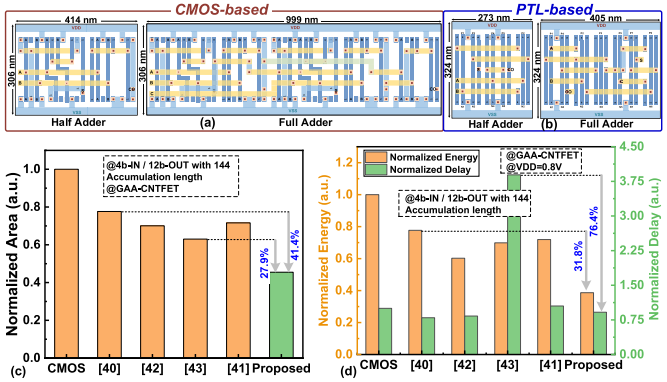


Fig. 15. The layouts of (a) CMOS-based and (b) the proposed PTL-based HA and FA (c) Area comparison for various adder trees. (d) Normalized energy and delay comparison of adder trees utilized in CIM macros.

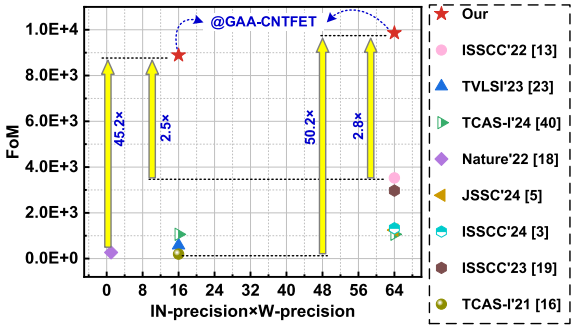


Fig. 16. FoM comparison with state-of-the-art CIM macros.

Table III compares the proposed SP-CIM with the state-of-the-art CIM macros. In [13], while the BiBL decryption strategy reduces some decryption delay, this work is limited to near-memory computing (NMC), where only a row of data can be activated for calculation, resulting in relatively low throughput of only 0.26 TOPS for a 4Mb array. Similarly, [19] also achieves only NMC. In [23], Chen et al. implemented in-memory self-decryption using SRAM, with multi-bit weights realized through a capacitance array. Expanding to higher bit precision would require exponentially larger capacitance values, leading to substantial area overhead. As a result, this work supports only 4b-IN and 4b-W. Additionally, while full-precision output can theoretically reach 13b, the output precision was quantized to 5b to reduce ADC overhead, causing a loss in computing accuracy and limiting its applicability in CNNs that demand high precision. In [16], Zhang et al. performed Boolean logic operations in the time domain, but the need for precise sampling of continuous signals complicates circuit implementation. Furthermore, due to the limited TMR of the MTJ, single-bit dot-product calculations in this design may exhibit a certain error rate. In [3] and [5], local approximate computing is employed to reduce area overhead. However, the errors produced in the aforementioned process may be further amplified during the shifting and accumulation processes in high-precision computations [6], [24].

In analog CIM macros, lower-precision ADCs are typically used to quantize the results, leading to outputs that are not at full precision. Therefore, to enable a fair comparison across different input and weight precisions, as well as between digital and analog CIM architectures, we used the figure-of-merit (FoM) as defined in [1] (energy efficiency × IN-precision × W-precision × out ratio). Due to the use of an efficient PTL-based adder tree, the proposed SP-CIM achieves FoMs of 8888 (calculated as 555.5 TOPS/W × 4 × 4 × 1) and 9862.4 at 4b-IN/4b-W/16b-OUT and 8b-IN/8b-W/24b-OUT, respectively, as shown in **Fig. 16**. The FoM of the SP-CIM architecture is increased by 45.2 and 2.5 times compared to [13] and [18], respectively at 4b-IN/4b-W/16b-OUT. The improvement in FoM is 50.2 and 2.8 times compared to those reported in [13] and [16] with 8b-IN/8b-W/24b-OUT.

E. Discussing

We have successfully demonstrated the first monolithic 3D integration of SOT-MRAM and CNTFET, as illustrated in **Fig. 17**. The integration scheme comprises two distinct functional layers: (1) The bottom SOT-MTJ layer was fabricated on a standard 8-inch silicon wafer through CMOS-compatible processes. (2) The upper CNTFET layer was subsequently integrated atop the SOT-MTJ structure using low-temperature processes. Since SOT-MRAM and CNTFET are compatible with back-end-of-line processes, their integration enables multi-layer 3D integration. This means that multiple layers of CNTFETs can be fabricated on top of the SOT-MTJ

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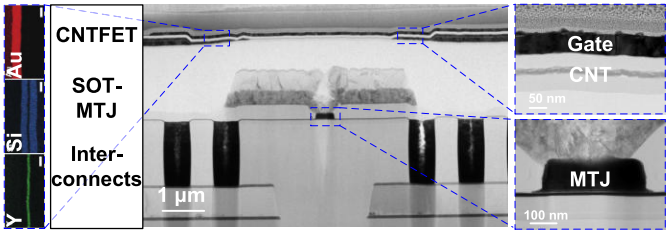


Fig. 17. Cross-sectional transmission electron microscope image of the integration of SOT-MRAM and CNTFET.

layer, allowing for ultra-high-density CNTFET-based MRAM. We will present CNTFET-based MRAM in detail in our future work. Therefore, the combination of SOT-MRAM and CNTFET for CIM macros holds great promise for extending the progress of Moore's Law.

V. Conclusion

In this study, a 288 kb SP-CIM macro was proposed utilizing VGSOT-MTJ and VGG-CNTFET technologies. The proposed SP-CIM macro successfully enables simultaneous data access, decryption, and full-accuracy MAC operations through a novel VSDC, eliminating the need for additional decryption logic. This design achieves significant improvements over existing in-memory decryption strategies, with a 45.7% reduction in energy consumption and an 87.2% reduction in delay. To further optimize area efficiency and minimize computing latency, a PTL-MC was developed, enabling local 2b-IN $\times$ 2b-W operations with only 20 transistors. Moreover, the implementation of PTL-based 10T-HA and 14T-FA in the local adder tree—supporting 4b-IN and 144 accumulation lengths—resulted in 31.8%, 76.4%, and 41.4% reductions in energy, delay, and area, respectively, compared to traditional adder trees in CIM macros. Simulations of the SP-CIM macro demonstrated the effectiveness of the proposed scheme, achieving a delay of 3.1 ns, throughput between 2.25–2.97 TOPS, and energy efficiency of 154.1–226.6 TOPS/W, while performing full-accuracy self-decryption MAC operations with 8b-IN/8b-W/24b-OUT. The FoM of the SP-CIM architecture is increased by 2.5–50.2 times compared to existing CIM macros. This work significantly enhances the practicality of using MRAM-CIM for secure AI edge computing.

References

[1] P.-C. Wu et al., "A 22nm 832Kb hybrid-domain floating-point SRAM in-memory-compute macro with 16.2-70.2TFLOPS/W for high-accuracy AI-edge devices," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2023, pp. 126–128.

[2] W. Zhang et al., "Edge learning using a fully integrated neuro-inspired memristor chip," *Science*, vol. 381, no. 6663, pp. 1205–1211, Sep. 2023.

[3] A. Guo et al., "34.3 a 22nm 64kb lightning-like hybrid computing-in-memory macro with a compressed adder tree and analog-storage quantizers for transformer and CNNs," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2024, pp. 570–572.

[4] N. Yang, X. Wang, X. Lin, and W. Zhao, "Exploiting carbon nanotube FET and magnetic tunneling junction for near-memory-computing paradigm," *IEEE Trans. Electron Devices*, vol. 68, no. 4, pp. 1975–1979, Apr. 2021.

[5] A. Guo et al., "A 28-nm 64-kb 31.6-TFLOPS/W digital-domain floating-point-computing-unit and double-bit 6T-SRAM computing-in-memory macro for floating-point CNNs," *IEEE J. Solid-State Circuits*, vol. 59, no. 9, pp. 3032–3044, Sep. 2024.

[6] H. Zhang et al., "SSM-CIM: An efficient CIM macro featuring single-step multi-bit MAC computation for CNN edge inference," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 70, no. 11, pp. 4357–4368, Nov. 2023.

[7] J. Luo et al., "Novel ferroelectric tunnel FinFET based encryption-embedded computing-in-memory for secure AI with high area-and energy-efficiency," in *IEDM Tech. Dig.*, Dec. 2022, pp. 36.5.1–36.5.4.

[8] M. Ashok, S. Maji, X. Zhang, J. Cohn, and A. P. Chandrakasan, "A secure digital in-memory compute (IMC) macro with protections for side-channel and bus probing attacks," in *Proc. IEEE Custom Integr. Circuits Conf. (CICC)*, Apr. 2024, pp. 1–2.

[9] Z. Tong et al., "A high throughput in-MRAM-computing scheme using hybrid p-SOT-MTJ/GAA-CNTFET," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 2, pp. 606–619, Feb. 2024.

[10] C. Wang, Z. Wang, Z. Zhang, Y. Zhang, and W. Zhao, "Area and energy efficient short-circuit-logic-based STT-MRAM crossbar array for binary neural networks," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 71, no. 3, pp. 1386–1390, Mar. 2024.

[11] C. Wang, Z. Wang, S. Li, Z. Zhang, and Y. Zhang, "Variation aware evaluation approach and design methodology for SOT-MRAM," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 4, pp. 1651–1664, Apr. 2024.

[12] H. Cai et al., "33.4 a 28nm 2Mb STT-MRAM computing-in-memory macro with a refined bit-cell and 22.4–41.5TOPS/W for AI inference," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2023, pp. 500–502.

[13] Y.-C. Chiu et al., "A 22nm 4Mb STT-MRAM data-encrypted near-memory computation macro with a 192GB/s read-and-decryption bandwidth and 25.1-55.1TOPS/W 8b MAC for AI operations," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, vol. 65, Feb. 2022, pp. 178–180.

[14] J. Wang et al., "RSACIM: Resistance summation analog computing in memory with accuracy optimization scheme based on MRAM," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 3, pp. 1014–1024, Mar. 2024.

[15] S. K. Roy et al., "Compute SNDR-boosted 22-nm MRAM-based in-memory computing macro using statistical error compensation," *IEEE J. Solid-State Circuits*, vol. 60, no. 3, pp. 1092–1102, Mar. 2025.

[16] Y. Zhang et al., "Time-domain computing in memory using spintronics for energy-efficient convolutional neural network," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 68, no. 3, pp. 1193–1205, Mar. 2021.

[17] J. Wang et al., "Reconfigurable bit-serial operation using toggle SOT-MRAM for high-performance computing in memory architecture," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 69, no. 11, pp. 4535–4545, Nov. 2022.

[18] S. Jung et al., "A crossbar array of magnetoresistive memory devices for in-memory computing," *Nature*, vol. 601, no. 7892, pp. 211–216, Jan. 2022.

[19] Y.-C. Chiu et al., "A 22nm 8Mb STT-MRAM near-memory-computing macro with 8b-precision and 46.4-160.1TOPS/W for edge-AI devices," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2023, pp. 496–498.

[20] Y. Sun et al., "CREAM: Computing in ReRAM-assisted energy-and area-efficient SRAM for reliable neural network acceleration," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 70, no. 8, pp. 3198–3211, Aug. 2023.

[21] S. Huang, H. Jiang, X. Peng, W. Li, and S. Yu, "XOR-CIM: Compute-in-memory SRAM architecture with embedded XOR encryption," in *Proc. 39th Int. Conf. Comput.-Aided Design*, Nov. 2020, pp. 1–6.

[22] T.-C. Chang et al., "13.4 a 22nm 1Mb 1024b-read and near-memory-computing dual-mode STT-MRAM macro with 42.6GB/s read bandwidth for security-aware mobile devices," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2020, pp. 224–226.

[23] Z. Chen et al., "PUF-CIM: SRAM-based compute-in-memory with zero bit-error-rate physical unclonable function for lightweight secure edge computing," *IEEE Trans. Very Large Scale Integr. VLSI Syst.*, vol. 31, no. 8, pp. 1234–1247, Aug. 2023.

[24] Z. Tong et al., "BSTCIM: A balanced symmetry ternary fully digital in-MRAM computing macro for energy efficiency neural network," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 12, pp. 6114–6127, Dec. 2024.

[25] Z. Tong et al., "In-MRAM computing based on complementary-sensing time-based readout circuit using hybrid VGSOT-MTJ/GAA-CNTFET," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 72, no. 1, pp. 173–177, Jan. 2025.

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[26] M. T. Nasab, A. Amirany, M. H. Moaiyeri, and K. Jafari, "High-performance and robust spintronic/CNTFET-based binarized neural network hardware accelerator," *IEEE Trans. Emerg. Topics Comput.*, vol. 11, no. 2, pp. 527–533, Apr. 2023.

[27] C.-S. Lee, E. Pop, A. D. Franklin, W. Haensch, and H. P. Wong, "A compact virtual-source model for carbon nanotube FETs in the sub-10-nm regime—Part II: Extrinsic elements, performance assessment, and design optimization," *IEEE Trans. Electron Devices*, vol. 62, no. 9, pp. 3070–3078, Sep. 2015.

[28] C.-S. Lee, E. Pop, A. D. Franklin, W. Haensch, and H.-S. P. Wong, "A compact virtual-source model for carbon nanotube FETs in the sub-10-nm regime—Part I: Intrinsic elements," *IEEE Trans. Electron Devices*, vol. 62, no. 9, pp. 3061–3069, Sep. 2015.

[29] K. Zhang, D. Zhang, C. Wang, L. Zeng, Y. Wang, and W. Zhao, "Compact modeling and analysis of voltage-gated spin-orbit torque magnetic tunnel junction," *IEEE Access*, vol. 8, pp. 50792–50800, 2020.

[30] M. M. Shulaker et al., "Three-dimensional integration of nanotechnologies for computing and data storage on a single chip," *Nature*, vol. 547, no. 7661, pp. 74–78, Jul. 2017.

[31] N. Yang, Z. Si, X. Wang, X. Lin, and W. Zhao, "Neuromorphic terahertz imaging based on carbon nanotube circuits," *Sci. China Inf. Sci.*, vol. 66, no. 6, Jun. 2023, Art. no. 169401.

[32] B. Wang et al., "Ionization and displacement damage on nanostructure of spin–orbit torque magnetic tunnel junction," *IEEE Trans. Nucl. Sci.*, vol. 69, no. 1, pp. 43–49, Jan. 2022.

[33] M. Zhu et al., "Radiation-hardened and repairable integrated circuits based on carbon nanotube transistors with ion gel gates," *Nature Electron.*, vol. 3, no. 10, pp. 622–629, Aug. 2020.

[34] M. T. Nasab, A. Amirany, M. H. Moaiyeri, and K. Jafari, "Hybrid MTJ/CNTFET-based binary synapse and neuron for process-in-memory architecture," *IEEE Magn. Lett.*, vol. 14, pp. 1–5, 2023.

[35] M. Rezaei, A. Amirany, M. H. Moaiyeri, and K. Jafari, "A high-accuracy and low-power emerging technology-based associative memory," *IEEE Trans. Nanotechnol.*, vol. 23, pp. 293–298, 2024.

[36] F. Razi, M. Hossein Moaiyeri, R. Rajaei, and S. Mohammadi, "A variation-aware ternary spin-Hall assisted STT-RAM based on hybrid MTJ/GAA-CNTFET logic," *IEEE Trans. Nanotechnol.*, vol. 18, pp. 598–605, 2019.

[37] A. Amirany, K. Jafari, and M. H. Moaiyeri, "True random number generator for reliable hardware security modules based on a neuromorphic variation-tolerant spintronic structure," *IEEE Trans. Nanotechnol.*, vol. 19, pp. 784–791, 2020.

[38] S. Salavati, M. H. Moaiyeri, and K. Jafari, "Ultra-efficient nonvolatile approximate full-adder with spin-hall-assisted MTJ cells for in-memory computing applications," *IEEE Trans. Magn.*, vol. 57, no. 5, pp. 1–11, May 2021.

[39] A. Amirany, M. H. Moaiyeri, and K. Jafari, "Nonvolatile associative memory design based on spintronic synapses and CNTFET neurons," *IEEE Trans. Emerg. Topics Comput.*, vol. 10, no. 1, pp. 428–437, Jan./Mar. 2022.

[40] W. Yi et al., "RDCIM: RISC-V supported full-digital computing-in-memory processor with high energy efficiency and low area overhead," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 4, pp. 1719–1732, Apr. 2024.

[41] H. Fujiwara et al., "A 5-nm 254-TOPS/W 221-TOPS/mm^2 fully-digital computing-in-memory macro supporting wide-range dynamic-voltage-frequency scaling and simultaneous MAC and write operations," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2022, pp. 1–3.

[42] Y.-D. Chih et al., "16.4 an 89TOPS/W and 16.3TOPS/mm^2 all-digital SRAM-based full-precision compute-in memory macro in 22nm for machine-learning edge applications," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, vol. 64, Feb. 2021, pp. 252–254.

[43] D. Wang, C. T. Lin, G. K. Chen, P. Knag, R. K. Krishnamurthy, and M. Seok, "DIMC: 2219TOPS/W 2569F2/b digital in-memory computing macro in 28nm based on approximate arithmetic hardware," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2022, pp. 266–268.

[44] W. Pan, Y. Yu, C. Tang, N. Yin, and Z. Yu, "Low-power, high-speed, and area-efficient multiplier based on the PTL logic style," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 71, no. 7, pp. 3538–3542, Jul. 2024.

[45] M. Vesterbacka, "A 14-transistor CMOS full adder with full voltage-swing nodes," in *Proc. IEEE Workshop Signal Process. Syst. Design Implement. (SiPS)*, Oct. 1999, pp. 713–722.

[46] N. Yin, W. Pan, Y. Yu, C. Tang, and Z. Yu, "Low-power pass-transistor logic-based full adder and 8-bit multiplier," *Electronics*, vol. 12, no. 15, p. 3209, 2023.

[47] P. Kumar, N. S. Bhandari, L. Bhargav, R. Rathi, and S. C. Yadav, "Design of low power and area efficient half adder using pass transistor and comparison of various performance parameters," in *Proc. Int. Conf. Comput., Commun. Autom. (ICCCA)*, May 2017, pp. 1477–1482.

[48] Y. Zhang et al., "Compact modeling of perpendicular-anisotropy CoFeB/MgO magnetic tunnel junctions," *IEEE Trans. Electron Devices*, vol. 59, no. 3, pp. 819–826, Mar. 2012.

[49] Z. Wang et al., "Evaluation of ultrahigh-speed magnetic memories using field-free spin–orbit torque," *IEEE Trans. Magn.*, vol. 54, no. 11, pp. 1–5, Nov. 2018.

[50] Z. Wang, W. Zhao, E. Deng, J.-O. Klein, and C. Chappert, "Perpendicular-anisotropy magnetic tunnel junction switched by spin-Hall-assisted spin-transfer torque," *J. Phys. D, Appl. Phys.*, vol. 48, no. 6, Feb. 2015, Art. no. 065001.

[51] C. Jiang et al., "Demonstration of a manufacturable SOT-MRAM multiplexer array towards industrial applications," *J. Semiconductors*, vol. 44, no. 12, Dec. 2023, Art. no. 122501.

[52] X. Dong, C. Xu, Y. Xie, and N. P. Jouppi, "NVSim: A circuit-level performance, energy, and area model for emerging nonvolatile memory," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 31, no. 7, pp. 994–1007, Jul. 2012.



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